

Question:

Melanie is a door-to-door saleswoman. She sold a third of her vacuum cleaners at the green house, 2 more to the red house, and half of what was left at the orange house. If Melanie has 5 vacuum cleaners left, how many did she start with?

Reasoning + Answer:

<think>
She sold a third at the green house , so the remaining two thirds represent $5 + 2 = 7$ vacuum cleaners .
If $2/3$ is 7 , then $1/3$ is 3.5 because $7 / 2 = 3.5$. This means she sold 3.5 vacuum cleaners at the green house . She sold 2 more to the red house , so she sold $3.5 + 2 = 5.5$. She sold half of what was left to the orange house , so 5.5 is half of what she started with at the orange house . Therefore , she started with $5.5 * 2 = 11$ vacuum cleaners at the orange house . She sold $5.5 + 2 = 7.5$ vacuum cleaners to the green and red houses , so she started with $11 + 7.5 = 18.5$. Since she can't sell half a vacuum cleaner , we round up to the nearest whole number , 19 .

</think>
<answer>
Melanie started with 19 vacuum cleaners .
</answer>

